

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2– Case Study (2 QUESTIONS)

Course Code:	ITA05	Course Title:	COMPUTER VISION
CO Assessed:	CO3 – Precision (BL4)	Assessment:	Assessment Tool 1 – Case Study
Weightage:	40% of CIA	Total Marks:	20 (2 Questions × 10Marks)
CO1 Statement:	<i>Analyze and extract image features using detection and matching techniques for effective image representation and comparison..(BL4)</i>		
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Section / Batch:	CSE AI	Date:	

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Code of Conduct

I G chandra sekhar reddy(192472173) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G chandra sekhar reddy

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Feature Detection and Matching	Preprocessing, Edge Detection, Harris Corner Detection, Hough Transform, SIFT	1	10
Feature Detection and Matching	PCA, Image Representation, Feature Matching, Image Processing Applications	2	10
GRAND TOTAL:			20 Marks

Annexure A –Case Study

1.A remote sensing application is used to align satellite images of the same geographical region captured at different times for monitoring environmental changes. The system initially relies on edge detection to identify matching regions between the images; however, due to differences in illumination, image orientation, and background details, the matching process becomes unreliable and produces inaccurate alignment results. The development team decides to replace the edge-based approach with Harris Corner Detection to identify stable and distinctive feature points. Analyze the limitations of edge detection in this application and explain how Harris Corner Detection improves feature localization, matching accuracy, and robustness for image registration under varying imaging conditions.

2.An autonomous surveillance system continuously captures high-resolution color images to detect vehicles and pedestrians in real time. Processing the original color images directly increases computational complexity and slows down feature extraction, making it difficult to achieve the required processing speed. To improve system performance, the engineers consider converting the images into grayscale or binary representations before applying feature detection algorithms. Analyze the impact of different image representations on computational efficiency, edge detection, and feature extraction performance, and justify the most appropriate image representation for achieving accurate and reliable object detection in real-time computer vision applications.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Faculty In-charge
Signature: _____
Date: _____

Harris Corner Detection for Remote Sensing Image Registration

Introduction

Remote sensing is one of the most important technologies used for observing and monitoring the Earth's surface. Satellites capture images of the same geographical region at different times to study changes in land use, urban development, agriculture, forest cover, water resources, disaster management, and environmental conditions. Since these images are acquired at different times, from different viewing angles, and under varying atmospheric conditions, they often do not perfectly overlap. Therefore, image registration is required to align multiple satellite images into a common coordinate system.

Image registration is the process of matching two or more images of the same scene so that corresponding features appear at the same locations. Accurate registration is essential for applications such as change detection, image fusion, object tracking, environmental monitoring, and Geographic Information Systems (GIS). The accuracy of image registration largely depends on the quality of feature extraction techniques used to identify matching points.

Traditional edge detection methods, such as Sobel, Prewitt, and Canny edge detectors, identify object boundaries by detecting sudden changes in image intensity. Although edge detection is useful for identifying structures within an image, it often fails to provide reliable feature points for image registration. Factors such as illumination changes, image rotation, scaling, noise, and background complexity reduce the effectiveness of edge-based matching.

To overcome these limitations, Harris Corner Detection is widely used in image registration. Instead of detecting continuous edges, the Harris detector identifies distinctive corner points where intensity changes significantly in multiple directions. These corners are highly stable, repeatable, and suitable for establishing accurate correspondences between images. Consequently, Harris Corner Detection has become one of the most popular feature detection techniques in remote sensing image registration.

Limitations of Edge Detection

Although edge detection is an important image processing technique, it has several drawbacks when used for satellite image registration.

1. Sensitivity to Illumination Changes

Satellite images are captured under different weather conditions, seasons, and times of the day. Variations in sunlight, cloud cover, and atmospheric conditions change the brightness and contrast of images. Since edge detectors rely mainly on intensity gradients, these lighting variations can produce different edge maps for the same scene, making accurate matching difficult.

For example, a building captured on a sunny day may produce strong edges, while the same building photographed on a cloudy day may generate weaker edges.

2. Poor Feature Distinctiveness

Edges represent continuous boundaries rather than unique points. Many edges look similar throughout an image, making it difficult to identify corresponding locations between two images. This often leads to ambiguous feature matching.

For instance, long road boundaries or river banks may produce identical edge patterns, increasing the possibility of incorrect matches.

3. Sensitivity to Orientation Changes

Satellite images may be captured from different viewing angles or with slight rotations. The orientation of detected edges changes with image rotation, reducing the reliability of edge-based matching.

Even small rotational differences can cause edge detectors to identify different edge locations, decreasing registration accuracy.

4. Background Interference

Remote sensing images contain numerous natural and man-made objects such as forests, buildings, roads, rivers, mountains, and agricultural fields. These objects generate a large number of edges, many of which are irrelevant to the registration process.

As a result, selecting correct matching edges becomes difficult, increasing computational complexity and reducing matching accuracy.

5. Inaccurate Localization

Edges indicate boundaries rather than exact feature points. During image registration, precise point-to-point correspondence is required. Since an edge extends over many pixels, determining the exact matching location becomes challenging.

This limitation decreases the precision of geometric transformation estimation.

6. Noise Sensitivity

Satellite images are often affected by sensor noise, atmospheric disturbances, and compression artifacts. Noise creates false edges that do not correspond to real objects. These unwanted edges reduce the reliability of image matching and may produce incorrect registration results.

Harris Corner Detection

Harris Corner Detection is a feature detection algorithm developed by Chris Harris and Mike Stephens in 1988. It detects points where image intensity changes significantly in multiple directions simultaneously. Such points are called ****corners****.

Corners usually occur at intersections of roads, building corners, bridge joints, window edges, and other distinctive geographical structures. Since these features remain relatively stable under different imaging conditions, they provide reliable landmarks for image registration.

Unlike edge detectors that identify continuous boundaries, the Harris detector focuses on localized feature points that are highly distinctive and easier to match.

Working Principle of Harris Corner Detection

The Harris Corner Detector analyzes a small window surrounding every pixel in the image.

The algorithm performs the following steps:

Step 1: Image Gradient Calculation

The intensity gradients in both horizontal and vertical directions are calculated.

- Horizontal gradient (I_x)
- Vertical gradient (I_y)

These gradients indicate how rapidly pixel intensity changes in different directions.

Step 2: Structure Tensor Formation

The local intensity information is summarized into a matrix called the **Structure Tensor** or **Second Moment Matrix**.

This matrix represents the intensity variation within a small neighborhood around each pixel.

Step 3: Corner Response Calculation

A mathematical equation computes the corner response value (R) for each pixel.

The response indicates whether the pixel represents:

- A flat region
- An edge
- A corner

Interpretation:

- **Large Positive R :** Strong corner
- **Small R :** Flat region
- **Negative R :** Edge

Step 4: Thresholding

Only pixels with corner response values greater than a predefined threshold are selected as candidate corners.

This eliminates weak and insignificant feature points.

Step 5: Non-Maximum Suppression

Multiple neighboring pixels may have high corner responses. Non-Maximum Suppression retains only the strongest corner within a local neighborhood.

This produces a set of well-localized, distinct feature points suitable for matching.

Advantages of Harris Corner Detection for Image Registration

1. Better Feature Localization

Corners represent precise image locations rather than long boundaries. Their exact coordinates can be determined accurately, improving registration precision.

2. High Repeatability

The same corners are usually detected even when images are captured under different lighting conditions or from slightly different viewpoints.

This consistency improves feature matching reliability.

3. Distinctive Feature Points

Corners are unique and easily distinguishable from surrounding image regions.

Examples include:

- Building corners
- Road intersections
- Railway crossings
- Bridge junctions
- Field boundaries

These features simplify correspondence between images.

4. Reduced False Matches

Since Harris detects only highly distinctive points, the number of incorrect matches is significantly reduced compared to edge-based methods.

This improves the overall accuracy of image registration.

5 . Robustness Against Noise

Although image noise affects all feature detectors to some extent, Harris Corner Detection is generally more robust than simple edge detection because it analyzes intensity variation over a local neighborhood rather than individual pixels.

6. Efficient Computation

The Harris algorithm is computationally efficient and suitable for processing large satellite images.

Its relatively low computational cost makes it practical for real-time and large-scale remote sensing applications.

7. Stable Under Small Transformations

Harris corners remain relatively stable under:

- Small image rotations
- Moderate illumination variations
- Small translations

This stability improves registration performance.

Applications in Remote Sensing

Harris Corner Detection is widely used in various remote sensing applications, including:

Image Registration

Matching corresponding feature points between multi-temporal satellite images.

Change Detection

Comparing registered images to identify urban growth, deforestation, flooding, and environmental changes.

Image Mosaicing

Combining multiple satellite images into one seamless large image.

GIS Mapping

Accurate alignment of satellite images with digital maps.

Disaster Monitoring

Comparing pre-disaster and post-disaster satellite images for damage assessment.

Urban Planning

Monitoring infrastructure development and land-use changes over time.

Comparison Between Edge Detection and Harris Corner Detection

Edge Detection	Harris Corner Detection
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Detects image boundaries	Detects distinctive corner points
Produces continuous edges	Produces localized feature points
Sensitive to illumination changes	More robust to lighting variations
Less suitable for feature matching	Highly suitable for feature matching
Difficult to localize exact points	Provides accurate point localization
Higher probability of false matches	Lower probability of false matches
Affected by background clutter	Focuses on important image features
Poor registration accuracy	High registration accuracy
Sensitive to noise	More robust against noise
Limited use in satellite image registration	Widely used in remote sensing applications

Conclusion

Image registration is a critical preprocessing step in remote sensing applications because it ensures that satellite images acquired at different times or from different sensors are accurately aligned. While traditional edge detection techniques are useful for identifying object boundaries, they suffer from several limitations such as sensitivity to illumination changes, orientation variations, noise, and background interference. Moreover, edges do not provide precise and distinctive points required for reliable feature matching.

Harris Corner Detection overcomes these challenges by identifying stable, unique, and well-localized corner points where image intensity changes significantly in multiple directions. These corner features remain relatively consistent across different imaging conditions and enable accurate correspondence between images. As a result, Harris Corner Detection significantly improves feature matching accuracy, reduces false matches, and enhances the overall quality of image registration.

Because of its robustness, computational efficiency, and reliable performance, Harris Corner Detection has become one of the most widely used feature detection methods in remote sensing. It plays an important role in applications such as environmental monitoring, change detection, GIS mapping, disaster assessment, urban planning, and satellite image mosaicing. Therefore, Harris Corner Detection is considered a highly effective and reliable technique for achieving precise remote sensing image registration.